

Data Requirements — Plants Module

Mango orchard tracking — Infineon / BIT Excellence Program
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1. Purpose

This document lists the data required to run the Plants module application, ahead of the technical design (database schema, backend, frontend). It does not cover implementation choices or sensor selection — only what information the application needs, independently of how it will be collected.

Reference values quoted where relevant (varieties, expected yields, investment figures) originate from the 2025 cohort's work (file sustainable_farm2025.xlsx, tab "Plants(Zalka)") and are reused as modifiable default values, not recalculated.

2. Scope

The Plants module covers six tasks assigned within the project:

- 1. Variety selection and monitoring
- 2. Planting calendar and growth stage tracking
- 3. Fertilizer supply and inventory management
- 4. Disease and pest surveillance
- 5. Maturity / harvest-readiness alerts
- 6. Nursery management (young plants before transplanting)

3. Data Requirements by Task

Each data field below has a unique reference ID (e.g. PL-04-01) for traceability across documents and discussions.

Task 1 — Variety Selection and Monitoring

Objective: Keep a clear record per planted variety, to track what was planted, where, and how it performs.

ID	Field	Description	Format	Mandatory	Possible source
PL-01-01	Variety name	Name of the mango variety (e.g. Kent, Keitt, Amélie)	Text	Yes	Manual entry — 2025 reference data
PL-01-02	Number of trees	Trees planted for this variety	Integer	Yes	Manual entry
PL-01-03	Spacing	Inter-row and intra-row spacing	Decimal (m)	Yes	Manual entry
PL-01-04	Expected yield	Reference yield used for planning	kg	Yes	Manual entry — 2025 reference data
PL-01-05	Tree vigor	General vigor assessment of the variety	Text	No	Manual entry
PL-01-06	Plot / block	Field block where the variety is planted	Text (block reference)	Yes	Manual entry
PL-01-07	Actual yield	Real yield recorded at harvest, compared against the expected yield	kg	No	Derived from harvest records (Task 5)

Task 2 — Planting Calendar and Growth Stage Tracking

Objective: Track when each block was planted and which growth stage it is currently in.

ID	Field	Description	Format	Mandatory	Possible source
PL-02-01	Planting date	Date the block was planted	Date	Yes	Manual entry
PL-02-02	Current stage	Nursery / growth / production	Status (enum)	Yes	Manual entry

ID	Field	Description	Format	Mandatory	Possible source
PL-02-03	Growth phase	0–2, 3–5, or 5–7 years bracket	Enum (derived)	Yes	Calculated from tree age
PL-02-04	Local rainfall	Rainfall data relevant to growth conditions	Decimal (mm)	No	External weather source or Energy module API
PL-02-05	Tree age	Age of the tree, computed automatically	Integer (calculated)	Yes	Calculated from planting date — never entered manually

Task 3 — Fertilizer Supply and Inventory Management

Objective: Keep fertilizer stock, usage, and replenishment under control per block.

ID	Field	Description	Format	Mandatory	Possible source
PL-03-01	Fertilizer type	Type/name of fertilizer	Text	Yes	Manual entry
PL-03-02	Stock quantity	Current quantity in stock	kg	Yes	Manual entry
PL-03-03	Minimum alert threshold	Stock level that triggers a reorder alert	kg	Yes	Manual entry
PL-03-04	Last restock date	Date of the last replenishment	Date	Yes	Manual entry
PL-03-05	Quantity applied per event	Amount used in one application, and the target block	kg + block reference	Yes	Manual entry
PL-03-06	Supplier	Supplier of the fertilizer batch	Text	No	Manual entry
PL-03-07	Person responsible	Who carried out the application	Text	No	Manual entry
PL-03-08	Application method	How the fertilizer was applied	Text	No	Manual entry

Task 4 — Disease and Pest Surveillance

Objective: Monitor tree health and detect problems early enough to act.

ID	Field	Description	Format	Mandatory	Possible source
PL-04-01	Health indicator	Overall tree health, 0–100% scale (displayed by category for field data-entry: severe stress / weakened / average / healthy / very healthy)	Percentage (stored) / Category (displayed)	Yes	Manual entry
PL-04-02	Detected issue	Disease or pest identified	Text	No	Manual entry, or automated detection via Fencing & Security module
PL-04-03	Observation date	Date the issue was observed	Date	Yes	Manual entry
PL-04-04	Block / tree affected	Location of the observed issue	Text (block reference)	Yes	Manual entry
PL-04-05	Treatment status	Follow-up status of the issue	Status (untreated / in progress / treated)	Yes	Manual entry
PL-04-06	Observer and method	Who observed it and how	Text	No	Manual entry

Task 5 — Maturity Alerts / Harvest Readiness

Objective: Know when fruit is ready to harvest and record what was actually picked.

ID	Field	Description	Format	Mandatory	Possible source
PL-05-01	Maturity status	Immature / ripe / ready for harvest	Status (enum)	Yes	Manual entry
PL-05-02	Estimated harvest date	Projected harvest date	Date	Yes	Derived from planting calendar (Task 2)
PL-05-03	Actual harvest date	Date the harvest actually took place	Date	Yes	Manual entry
PL-05-04	Harvested quantity	Quantity picked	kg	Yes	Manual entry
PL-05-05	Batch number	Unique identifier assigned at harvest, used for traceability by downstream modules	Unique ID	Yes	Generated by the Plants module at harvest time

Task 6 — Nursery Management

Objective: Track young plants from germination through to transplanting into the field.

ID	Field	Description	Format	Mandatory	Possible source
PL-06-01	Number of young plants	Plants currently in the nursery	Integer	Yes	Manual entry
PL-06-02	Development stage	Current stage of the young plant	Status (enum)	Yes	Manual entry
PL-06-03	Planned transplant date	Expected date of transfer to the field	Date	Yes	Manual entry
PL-06-04	Survival / loss rate	Share of plants that survive the nursery stage	Percentage	No	Manual entry
PL-06-05	Plant origin	External purchase or internal production	Text	No	Manual entry

4. Reading Notes

- Every field is recorded with its collection date and origin, so that the source of a data point (manual entry, sensor, or another module's API) can change later without altering the application's structure.
- "Plot/block" (PL-01-06, PL-03-05, PL-04-04) is the linking key across tasks — without it, an overall health score cannot distinguish a uniformly moderate orchard from a mix of severely stressed and fully healthy blocks.
- Numeric reference values (yields, investment figures) come from the 2025 cohort and are not being questioned at this stage.

5. References

sustainable_farm2025.xlsx — tab "Plants(Zalka)", 2025 cohort deliverable.